

RC1

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- Major in ECE, looking forward to be a PhD student.
- Interested in math and physics.
- Hobbies: reading, sleeping (yes I am a sleepyhead), coffee, travelling.
- Feel free to ask me anything about life, course selection, research, etc.

- Don't be frightened by "Honor" in the course title. Remember 156 is also an honor course.
- As you may be told by many, 186 focuses more on proofs and rigorous concepts, not calculations.
- It is not hard to get a good grade in 186, remember to attend lectures on time and finish assignments by yourself.
- Taking Notes is a good habit, personally.
- If you are worried about exams, make sure you finish assignments, sample exams and quizzes.

- Pays more attention to the concepts and the ideas behind theorems.
- Focuses on some important insights and connections between concepts.
- I am not a fan of heavy calculations, there will be some exercises to help you understand.
- Feel free to interrupt and ask any questions if you have any doubt.

- Statements: Can be judged true or false.
- Predicates / statement frame: Statements with variables.
- Quantifiers: $\forall, \exists$; Hanging Quantifiers.
- Logical Operations: $\neg, \wedge, \vee, \rightarrow$
- Truth Table.

Key Takeaway

Statements Frame + Quantifiers = Statements;
Single Statements + Logical Operations = Compound Statements.

Definition

- A Contradiction is a statement that is always false, regardless of the truth values of its components.
- A Tautology is a statement that is always true, regardless of the truth values of its components.
- Vacuous Truth: All pink pigs can fly. (Not a Tautology)

e.g.

- 1 $P \vee \neg P$ is a tautology.
- 2 $P \rightarrow P$ is a tautology.
- 3 $P \wedge \neg P$ is a contradiction.

- Implication: $P \rightarrow Q$.
- Converse: $Q \rightarrow P$.
- Inverse: $\neg P \rightarrow \neg Q$.
- Two statements P and Q are logically equivalent, denoted by $P \equiv Q$, if they have the same truth value in every possible scenario. Note: Their Equivalence is a Tautology.
- $P \rightarrow Q \equiv \neg P \vee Q$.
- Contrapositive: $\neg Q \rightarrow \neg P \equiv P \rightarrow Q$. **Important for Proof!**
- **De Morgan's Laws:**
 - $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.
 - $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$.
- **Distributive Laws:**
 - $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$.
 - $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$.

Prove some important theorem

$$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$$

Prove:

- 1 by using only logical identities
- 2 by directly using truth table

Hint: use $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$ and associative laws.

Compare two ways, what did you find?

Please write down the negation of the statement:

$$\forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n > N, |a_n - a| < \epsilon$$

3. Set Theory

- Define by Listing. $A = \{1, 2, 3\}$
- Define by Predicate. $B = \{x | x \text{ is an even number}\}$
- Set $A = B$ iff $A \subseteq B$ and $B \subseteq A$. (Important for Proof!)

- Union: $A \cup B = \{x | x \in A \text{ or } x \in B\}$
- Intersection: $A \cap B = \{x | x \in A \text{ and } x \in B\}$
- Difference: $A \setminus B = \{x | x \in A \text{ and } x \notin B\}$
- Complement: $\bar{A} \quad (A^C) = U \setminus A$

The laws for logical equivalencies immediately lead to several rules for set operations. For example, de Morgan's laws (see Example 1.1.11) for $\wedge$ and $\vee$ imply

- ▶ $(A \cup B)^c = A^c \cap B^c$,
- ▶ $(A \cap B)^c = A^c \cup B^c$,

Other such rules are, for example,

- ▶ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- ▶ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- ▶ $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$
- ▶ $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$
- ▶ $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
- ▶ $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

Remark: It is easier to understand by drawing Venn Diagram.

- Power Set: $\mathcal{P}(A) = \{X | X \subseteq A\}$.
- Cardinality: $|A|$ is the number of elements in set A .
- Finite Set: A set with a finite number of elements.
- Infinite Set: A set with an infinite number of elements.
- Countable Set: A set that can be put into one-to-one correspondence with the natural numbers.
- Uncountable Set: A set that cannot be put into one-to-one correspondence with the natural numbers.
- $|\mathcal{P}(A)| = 2^{|A|}$ if A is finite.

About measure theory, search for "Hilbert's Hotel Paradox"

<https://www.youtube.com/watch?v=0xGsU8oIWjY>

- Ordered Pair: (a, b) , where a is the first element and b is the second element.
- Two ordered pairs are equal iff their corresponding elements are equal. $(a, b) = (c, d) \Leftrightarrow a = c \text{ and } b = d$.
- Cartesian Product: $A \times B = \{(a, b) | a \in A, b \in B\}$.
- Note: $A \times B \neq B \times A$ in general.
- n-tuple: generalization of ordered pair.
- Construction of ordered pair and Natural Numbers:
 $(a, b) = \{\{a\}, \{a, b\}\}; 0 = \emptyset, 1 = \{\emptyset\}, 2 = \{\emptyset, \{\emptyset\}\}, \dots$

- Naive Set Theory: A set is any collection of distinct objects.
- Problems: Russell's Paradox. Let $R = \{x | x \notin x\}$, then $R \in R \Leftrightarrow R \notin R$. Not all predicates can define a set.
- Axiomatic Set Theory: Zermelo-Fraenkel Set Theory (ZF), ZF with Axiom of Choice (ZFC).
- More discussion: Godel's Incompleteness Theorems, etc.

- Axiom of Extensionality: Two sets are equal iff they have the same elements.
- Axiom of Regularity: Every non-empty set A contains an element that is disjoint from A .
- Axiom of Specification: For any set A and any property P , there is a subset of A containing exactly those elements of A that satisfy P .
- Axiom of Pairing: For any sets A and B , there is a set $\{A, B\}$ that contains exactly A and B as elements.
- Axiom of Union: For any set A , there is a set that contains exactly the elements of the elements of A .
- Axiom of Power Set: For any set A , there is a set $\mathcal{P}(A)$ that contains exactly the subsets of A .

- Axiom of Infinity: There exists a set that contains the empty set and is closed under the operation of adding a singleton.
(Construction of Natural Numbers)
- Axiom of Replacement: If A is a set and F is a definable function, then the image of A under F is also a set.
- Axiom of Choice: For any set A of non-empty sets, there exists a choice function f defined on A such that for every set B in A , $f(B)$ is an element of B .

Let A, B, C be three statements. Use truth table to prove that

$$(A \vee B) \wedge C \equiv (A \wedge C) \vee (B \wedge C)$$

Let A, B, C be three sets. Prove that

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

Venn Diagram?

4. Numbers: Natural, Rational, Real, Complex

- Natural Numbers: $\mathbb{N} = \{0, 1, 2, 3, \dots\}$.
- Properties w.r.t. Addition and Multiplication.
- Mathematical Induction: Basis Step, Inductive Step.

- The **Axiom of Infinity** allows us to construct the natural numbers from set theory.
- It asserts that there exists a set, let's call it I , which contains the empty set $\emptyset$ and for every set x in I , the set $x \cup \{x\}$ is also in I .
- We define the successor of a set x as $S(x) = x \cup \{x\}$.
- We can then define the natural numbers as follows:
 - $0 := \emptyset$
 - $1 := S(0) = 0 \cup \{0\} = \emptyset \cup \{\emptyset\} = \{\emptyset\}$
 - $2 := S(1) = 1 \cup \{1\} = \{\emptyset\} \cup \{\{\emptyset\}\} = \{\emptyset, \{\emptyset\}\}$
 - $3 := S(2) = 2 \cup \{2\} = \{\emptyset, \{\emptyset\}\} \cup \{\{\emptyset, \{\emptyset\}\}\} = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$
 - and so on...
- The set of natural numbers $\mathbb{N}$ is the smallest set satisfying the axiom of infinity.

- **Addition & Multiplication:**

- **Associativity:** For any $a, b, c \in \mathbb{N}$, we have $(a + b) + c = a + (b + c)$ and $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.
- **Existence of a neutral element:** For addition, there is an element 0 such that $a + 0 = a$ for all $a \in \mathbb{N}$. For multiplication, there is an element 1 such that $a \cdot 1 = a$ for all $a \in \mathbb{N}$.
- **Commutativity:** For any $a, b \in \mathbb{N}$, we have $a + b = b + a$ and $a \cdot b = b \cdot a$.

- **Distributivity:** Multiplication distributes over addition. For any $a, b, c \in \mathbb{N}$, $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$.
- **Ordering:** The natural numbers are well-ordered. For any two natural numbers a and b , exactly one of the following holds: $a < b$, $a = b$, or $a > b$.

Often one wants to show that some statement frame $A(n)$ is true for all $n \in \mathbb{N}$ with $n \geq n_0$ for some $n_0 \in \mathbb{N}$. Mathematical induction works by establishing two statements:

- (I) $A(n_0)$ is true.
- (II) $A(n + 1)$ is true whenever $A(n)$ is true for $n \geq n_0$, i.e.,

$$\forall_{n \in \mathbb{N}, n \geq n_0} (A(n) \Rightarrow A(n + 1))$$

Note that (II) does not make a statement on the situation when $A(n)$ is false; it is permitted for $A(n + 1)$ to be true even if $A(n)$ is false.

The **principle of mathematical induction** now claims that $A(n)$ is true for all $n \geq n_0$ if (I) and (II) are true. Essentially, this principle is part of the definition of $\mathbb{N}$.

Exercise: use induction to prove that for Fibonacci,

$$F_0 = 0, F_1 = 1, F_{n+2} = F_n + F_{n+1}$$

the formula is

$$F_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right]$$

We assume that the set of rational numbers (fractions) is known.

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z} \wedge q \neq 0 \right\}$$

Properties w.r.t Addition:

- Associativity
 - Commutativity
 - Neutral element
 - Inverse element
- Distributivity also holds.

Properties w.r.t Multiplication:

- Associativity
- Commutativity
- Neutral element
- Inverse element

We define the **absolute value** (sometimes called the **modulus**) $|a|$ of a rational number a by

$$|a| = \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

An important basic result is the **triangle inequality**:

Theorem (Triangle Inequality)

For all rational numbers $a, b \in \mathbb{Q}$, we have

$$|a + b| \leq |a| + |b|.$$

Inverse triangle inequality:

$$|a - b| \geq ||a| - |b||$$

We assume that we know what a strictly positive rational number is, i.e., that there exists a set P with the property that for every number a one and only one of the following holds:

- (i) $a = 0$,
- (ii) $a \in P$,
- (iii) $-a \in P$.

This is known as the **trichotomy law** for the rational numbers.

$$\text{associativity} \quad \forall_{a,b,c \in \mathbb{Q}} \quad a + (b + c) = (a + b) + c \quad (\text{P1})$$

$$\text{neutral element} \quad \exists_{0 \in \mathbb{Q}} \forall_{a \in \mathbb{Q}} \quad a + 0 = 0 + a = a \quad (\text{P2})$$

$$\text{inverse element} \quad \forall_{a \in \mathbb{Q}} \exists_{-a \in \mathbb{Q}} \quad (-a) + a = a + (-a) = 0 \quad (\text{P3})$$

$$\text{commutativity} \quad \forall_{a,b \in \mathbb{Q}} \quad a + b = b + a \quad (\text{P4})$$

$$\text{associativity} \quad \forall_{a,b,c \in \mathbb{Q}} \quad a \cdot (b \cdot c) = (a \cdot b) \cdot c \quad (\text{P5})$$

$$\text{neutral element} \quad \exists_{1 \in \mathbb{Q}, 1 \neq 0} \forall_{a \in \mathbb{Q}} \quad a \cdot 1 = 1 \cdot a = a \quad (\text{P6})$$

$$\text{inverse element} \quad \forall_{a \in \mathbb{Q}, a \neq 0} \exists_{a^{-1} \in \mathbb{Q}} \quad a \cdot a^{-1} = a^{-1} \cdot a = 1 \quad (\text{P7})$$

$$\text{commutativity} \quad \forall_{a,b \in \mathbb{Q}} \quad a \cdot b = b \cdot a \quad (\text{P8})$$

$$\text{distributivity} \quad \forall_{a,b,c \in \mathbb{Q}} \quad a \cdot (b + c) = a \cdot b + a \cdot c \quad (\text{P9})$$

Trichotomy Law (P10): There exists a set $P \subset \mathbb{Q}$ of positive rationals such that for every $a \in \mathbb{Q}$, one and only one of the following holds: (i) $a = 0$, (ii) $a \in P$, (iii) $-a \in P$.

We also assume that the set of positive numbers is closed under addition and multiplication:

- If a and b are in P , then $a + b$ is in P . (P11)
- If a and b are in P , then $a \cdot b$ is in P . (P12)

Prove that $\sqrt{2} + \sqrt{3}$ is not a rational number.

Definition

We say that a set $U \subset \mathbb{Q}$ is **bounded** if there exists a constant $c \in \mathbb{Q}$ such that

$$|x| \leq c \quad \text{for all } x \in U.$$

If this is not the case, we say that U is **unbounded**.

Numbers c_1 and c_2 such that

$$c_1 \leq x \leq c_2 \quad \text{for all } x \in U$$

are called **lower** and **upper bounds** for U , respectively.

We say that a set $U \subset \mathbb{Q}$ is **bounded above** if there exists an upper bound for U , and **bounded below** if there exists a lower bound for U .

Definition

Let $U \subset \mathbb{Q}$ be a subset of the rational numbers. We say that a number $x_1 \in U$ is the **minimum** of U if

$$x_1 \leq x, \quad \text{for all } x \in U$$

and we write $x_1 =: \min U$.

Similarly, we say that $x_2 \in U$ is the **maximum** of U if

$$x_2 \geq x, \quad \text{for all } x \in U$$

and write $x_2 =: \max U$.

Definition

We say that an upper bound $c_2 \in \mathbb{Q}$ of a set $U \subset \mathbb{Q}$ is the **least upper bound** or **supremum** of U if no $c \in \mathbb{Q}$ with $c < c_2$ is an upper bound. We then write $c_2 =: \sup U$.

We say that a lower bound $c_1 \in \mathbb{Q}$ is the **greatest lower bound** or **infimum** of U if no $c \in \mathbb{Q}$ with $c > c_1$ is a lower bound. We then

Definition

Let $A \subset \mathbb{R}$ be any set.

- (i) We call $x \in \mathbb{R}$ an **interior point of A** if there exists some $\varepsilon > 0$ such that the interval $(x - \varepsilon, x + \varepsilon) \subset A$. The set of interior points of A is denoted by $\text{int } A$.
- (ii) We call $x \in \mathbb{R}$ an **exterior point of A** if there exists some $\varepsilon > 0$ such that the interval $(x - \varepsilon, x + \varepsilon) \cap A = \emptyset$.
- (iii) We call $x \in \mathbb{R}$ a **boundary point of A** if for every $\varepsilon > 0$ $(x - \varepsilon, x + \varepsilon) \cap A \neq \emptyset$ and $(x - \varepsilon, x + \varepsilon) \cap A^c \neq \emptyset$. The set of boundary points of A is denoted by ∂A .
- (iv) We call $x \in \mathbb{R}$ an **accumulation point of A** if for every $\varepsilon > 0$ the interval $(x - \varepsilon, x + \varepsilon) \cap A \setminus \{x\} \neq \emptyset$.

Definition

- (i) A set $A \subset \mathbb{R}$ is called **open** if all points of A are interior points, i.e., if

$$A = \text{int } A.$$

- (ii) A set $A \subset \mathbb{R}$ is called **closed** if $\mathbb{R} \setminus A$ is open.

- (iii) The set

$$\overline{A} := A \cup \partial A$$

is called the **closure** of A . It can be shown that $\overline{A}$ is the smallest closed set that contains A .

Further, think about sets in $\mathbb{R}$ that are both open and closed.

Rational numbers are **insufficient**.

We extend real numbers to be algebraically complete by introducing imaginary number i . The complex numbers $\mathbb{C}$ is defined by using ordered pair:

$$\mathbb{C} := \{(a, b) : a, b \in \mathbb{R}\} = \mathbb{R}^2,$$

such that (a, b) can be denoted by $a + bi$. Just use your high school knowledge to do the calculations!

The modulus or absolute value of a complex number $z = a + bi$ is defined as:

$$|z| = \sqrt{a^2 + b^2} = \sqrt{z \cdot \bar{z}},$$

where $\bar{z} = a - bi$.

In fact, it is possible to prove that $\mathbb{C}$ is algebraically closed:

Theorem (Fundamental Theorem of Algebra)

For $a_0, \dots, a_n \in \mathbb{C}$, $a_n \neq 0$, the expression

$$p(x) = a_0 + a_1x + \cdots + a_nx^n$$

is said to be a **polynomial of degree n** . There exist numbers $\alpha, \lambda_1, \dots, \lambda_n \in \mathbb{C}$ such that

$$p(x) = \alpha \cdot (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n). \quad (1.6.1)$$

We say that the λ_k are the **roots** or **zeroes** of p . The representation (1.6.1) is called the **factorization** of $p(x)$.

If exactly m of the λ_k in (1.6.1) are equal to the same number $\lambda \in \mathbb{C}$, then m is said to be the **multiplicity** of the root λ .

- Vv186 Slides
- Vv186 RC, 24 Fall, ZhuoCheng He
- Vv186 RC, 24 Fall, Ruijie Zheng

Thank you!